

JEE MAIN REVISION MASTER: WORK POWER & ENERGY

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 153 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Work Power And Energy - Core Application	153 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Work Power And Energy - Core Application (Very High Yield — 153 PYQs)

Core Principle: Key problem pattern in Work Power And Energy.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A body of mass 1 kg moves along a straight line with a velocity $v=2x^2$. The work done by the body during displacement from $x=0$ to 5 m..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "A spherical ball of mass 2 kg falls from a height of 10 m and is brought to rest after penetrating 10 cm into sand. The average force exerted..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : An object moves from position $r_1(1)$ to position $r_1(2)$ under a conservative force field ϕ ..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

A body of mass 1 kg moves along a straight line with a velocity $v=2x^2$. The work done by the body during displacement from $x=0$ to 5 m is ____ J.

- (A) 0
- (B) 250
- (C) 1250 [Correct]
- (D) 1000

Correct Answer: C

Solution Analysis: Work done (W) is defined as the integral of force (F) over a displacement (dx): $W = \int_{x_i}^{x_f} F dx$. From Newton's Second Law, $F=ma$, and acceleration $a=(dv/dt)$. Using the chain rule, $a=(dv/dx) \cdot (dx/dt)=v(dv/dx)$. $W = \int_{x_i}^{x_f} [m(v(dv/dx))] dx \Rightarrow W = \int_{v_i}^{v_f} mv dv = m \int_{v_i}^{v_f} v dv = m [\frac{v^2}{2}]_{v_i}^{v_f} \Rightarrow W = (\frac{1}{2})mv_f^2 - (\frac{1}{2})mv_i^2 \Rightarrow W = \Delta \text{ K.E.}$ The mass of the body is $m=1\text{kg}$. The velocity of body is given as function of position, $v=2x^2$. Init